Team ID	Project Title	Participants	Comments
1	Matrix Factorization, SVD, Funk SVD, Gradient Descent	**Onur Boyar** Ramazan Yarar Oğuz Kaplan Emre Yüksel	Building recommender systems without using a library to do matrix computations and optimizations. We will try different approaches and compare them in terms of their performance. **ACM RecSys Challenge 2020**
2	User Engagement Prediction	**Uğur Ali Kaplan** Metehan Seyran Janberk Sü Elif Sargut	User engagement prediction is a subfield of recommender systems, since companies such as Twitter would like to show you the tweets you are most likely to engage with. The goal is to predict the probability for different types of engagement (Favourite, Reply, Retweet and Quote Tweet) of a target user for a set of tweets, based on heterogeneous input data. This is the **ACM RecSys2020 Challenge**. Details can be read from http://www.recsyschallenge.com/2020/ and if you decide to participate in the challenge, you also have to agree to terms & conditions at http://www.recsyschallenge.com/2020/Terms%20and%20Conditions%20in%20the%20RecSys%20Challenge%202020.pdf There are many different methods we could explore. It is possible to use matrix factorization methods, various deep learning algorithms and even reinforcement learning. We can decide on which methods to try after discussing our strong and weak points. We can also rapidly prototype different methods and focus on the most promising, say 2 methods and improve upon them. If we actually end up being successful (Top 5), we will be expected to **write a workshop paper for ACM RecS**ys 2020 Conference.
3	Conversational Information Seeking (CIS).	**Ulaş Mergen** Can Bulguoğlu Büşra Gökmen Ceren Yağmur Dardağan	There are currently few datasets appropriate for training and evaluating models for Conversational Information Seeking (CIS). The main aim of TREC CAsT is to advance research on conversational search systems. The goal of the track is to create a reusable benchmark for open-domain information centric conversational dialogues. The track will run in 2019 and establish a concrete and standard collection of data with information needs to make systems directly comparable. This is the first year of TREC CAsT, which will run as a track in TREC. This year we aim to focus on candidate information ranking in context: Read the dialogue context: Track the evolution of the information need in the conversation, identifying salient information needed for the current turn in the conversation Retrieve Candidate Response Information: Perform retrieval over a large collection of paragraphs (or knowledge base content) to identify relevant information
4	Robustness of Computer Vision Systems	**Batuhan Faik Derinbay** Atahan Özer Ensar Buğrahan Erdap Mehmet Ali Özer	*Widely available, large datasets *Various pre-made basis models with open source code *Many papers on each dataset and different robustness techniques available for many *There is a suitable upcoming vision challenge in April, due August -Winners gets to present their work in an ECCV 2020 workshop -Challenge link: http://www.robustvision.net/
5	Anomaly detection using graph neural networks, graph convolutional networks	**Güneykan Özgül** Onur Kılıçoğlu Berk Sudan	Graph Neural Network are used for anomaly detection, image and text classification and combinatorial optimization problem. There are many approaches about detecting anomalies in both unsupervised and semi-supervised learning on graph-structured data . We can reproduce the results of the papers and increase the model score by creating new hypotheses and apply them to different datasets.
6	Traffic Anomaly Detection	**Uras Mutlu** Halil Eralp Koçaş Abdullah Baş Can Bulguoğlu	AI City Challenge Track #4: Detecting anomalies in traffic, including wrong turns, wrong driving direction, lane change errors, and all other anomalies, based on video feeds available from multiple cameras at intersections and along highways.